restart

Calculer la densité de porteurs créés dans l'eau à 800 nm, en fonction de l'intensité du laser.

Rapporter à l'intensité incidente dans l'air.

Modèle d'ionisation : Keldysh

$$Ionisation := t \rightarrow \frac{2.\cdot\omega_l}{9.\cdot\pi}\cdot\left(\frac{\omega_l\cdot me}{\hbar\cdot Keldysh_1}\right)^{\frac{3.}{2.}}\cdot KeldyshFunction\cdot\exp\left(-\pi\cdot\mathrm{trunc}\left(\frac{U_{eff}}{\hbar\cdot\omega_l}+1.\right)\cdot\left(\frac{(\mathrm{EllipticK}(Keldysh_1)-\mathrm{EllipticE}(Keldysh_1))}{\mathrm{EllipticE}(Keldysh_2)}\right)\right):$$

$$KeldyshFunction := \mathrm{sqrt}\left(\frac{\pi}{2.\cdot\mathrm{EllipticK}(Keldysh_2)}\right)\cdot sum\left(\exp\left(-\frac{\pi\cdot n\cdot(\mathrm{EllipticK}(Keldysh_1)-\mathrm{EllipticE}(Keldysh_1))}{\mathrm{EllipticE}(Keldysh_2)}\right)\cdot KeldyshFunction2\left(\pi\cdot\mathrm{sqrt}\left(\frac{\left(2.\cdot\mathrm{trunc}\left(\frac{U_{eff}}{\hbar\cdot\omega_l}+1.\right)-\frac{2.\cdot U_{eff}}{\hbar\cdot\omega_l}+n\right)}{2.\cdot\mathrm{EllipticK}(Keldysh_2)\cdot\mathrm{EllipticE}(Keldysh_2)}\right)\right), n=0..\infty\right):$$

$$KeldyshFunction2 := z \rightarrow int(\exp(y^{2.}-z^{2.}), y=0..z):$$

$$U_{eff} := \frac{2.\cdot U_g}{\pi\cdot Keldysh_1}\cdot\mathrm{EllipticE}(Keldysh_2):$$

$$Keldysh_1 := \frac{Keldysh_0}{\mathrm{sqrt}(1.+Keldysh_0^{2.})};\ Keldysh_2 := \frac{Keldysh_1}{Keldysh_0};\ Keldysh_0 := \frac{\omega_l\cdot\mathrm{sqrt}(me\cdot U_g)}{ec\cdot\mathrm{sqrt}\left(\left(\frac{2.\cdot Iabs(t)}{c\cdot\varepsilon 0}\right)\right)};$$

$$\frac{Keldysh_0}{\sqrt{1.+Keldysh_0^{2.}}}$$

$$\frac{1}{\sqrt{1.+Keldysh_0^{2.}}}$$

$$\frac{0.7071067814\,\omega_l\sqrt{me\,U_g}}{ec\sqrt{\frac{Iabs(t)}{c\,\varepsilon 0}}} \tag{1}$$

$$U_g := 6.5\cdot ec;\ ec := 1.6e-19:\ \omega_l := \frac{2.\cdot\mathrm{Pi}\cdot c}{\mathrm{lambda}};\ \mathrm{lambda} := 800E-9;\ c := 3E8;\ \hbar := \frac{h}{2.\cdot\mathrm{Pi}}:\ h$$

$:= 6.63e{-}34;\ epsilon0 := 8.85e{-}12;\ me := 9.1e{-}31 : \mathrm{tau} := 100e{-}15 : I0 := 1e16$

$$6.5\ ec$$

$$\frac{2.\ \pi\ c}{\lambda}$$

$$8.00\ 10^{-7}$$

$$3.\ 10^{8}$$

$$6.63\ 10^{-34}$$

$$8.85\ 10^{-12}$$

$$1.\ 10^{16} \qquad \textbf{(2)}$$

Le modèle est trop complexe pour pouvoir être résolu proprement. Voyons celui de Vogel.

$restart : Digits := 20 :$

$$MPI := t \rightarrow \frac{2e0 \cdot \mathrm{omega}}{9e0 \cdot \mathrm{Pi}} \cdot \left(\frac{0.5e0 \cdot m \cdot \mathrm{omega}}{\hbar} \right)^{\frac{3e0}{2e0}} \cdot \left(\frac{ec^2}{16e0 \cdot 0.5e0 \cdot m \cdot Egap \cdot \omega^2 \cdot c \cdot epsilon0 \cdot n} \cdot Iabs(t) \right)^k \cdot \exp(2 \cdot k) \cdot \mathrm{phi}\left(\mathrm{sqrt}\left(2e0 \cdot k - \frac{2 \cdot Egap}{\hbar \cdot \mathrm{omega}} \right) \right);\ \mathrm{phi} := x \rightarrow \exp(-x^2) \cdot int(\exp(y^2),\ y = 0\,..x)$$

$$t \rightarrow \frac{2.\ \omega \left(\frac{0.5\ m\ \omega}{\hbar} \right)^{\frac{3.}{2.}} \left(\frac{ec^2\ Iabs(t)}{16. \cdot 0.5\ m\ Egap\ \omega^2\ c\ \varepsilon 0\ n} \right)^k \mathrm{e}^{2\,k}\ \phi\left(\sqrt{2.\ k - \frac{2\ Egap}{\hbar\ \omega}} \right)}{9.\ \pi}$$

$$x \rightarrow \mathrm{e}^{-x^2} \left(\int_0^x \mathrm{e}^{y^2}\, \mathrm{d}y \right) \qquad \textbf{(3)}$$

$$k := \mathrm{trunc}\left(\frac{Egap}{\hbar \cdot \mathrm{omega}} + 1 \right)$$

$$\mathrm{trunc}\left(\frac{Egap}{\hbar\ \omega} + 1 \right) \qquad \textbf{(4)}$$

$$Iabs := t \rightarrow I0 \cdot \exp\left(-4. \cdot \ln(2.) \cdot \left(\frac{t}{\tau_{laser}} \right)^{2.} \right)$$

$$t \rightarrow I0\ \mathrm{e}^{(-1) \cdot 4. \ln(2.) \left(\frac{t}{\tau_{laser}} \right)^{2.}} \qquad \textbf{(5)}$$

$Egap := 6.5 \cdot ec;\ ec := 1.6e{-}19 : \mathrm{omega} := \frac{2. \cdot \mathrm{Pi} \cdot c}{\mathrm{lambda}};\ \mathrm{lambda} := 800e{-}9;\ c := 3E8;\ \hbar := \frac{h}{2. \cdot \mathrm{Pi}} : h := 6.63e{-}34;\ epsilon0 := 8.85e{-}12;\ m := 9.1e{-}31 : M := 3e{-}26 : \tau_{coll} := 1e{-}15 : \tau_{laser} := 30e{-}15 :$

$$6.5\ ec$$

$$\frac{2.\ \pi\ c}{\lambda}$$

$$8.00\ 10^{-7}$$

$$3.\ 10^{8}$$

$$6.63\ 10^{-34}$$

$$8.85\ 10^{-12} \quad \textbf{(6)}$$

$$R := 0.5039\mathrm{e}{-2};\ n := 1.329;\ I0 := evalf\left(\frac{F}{\tau_{laser}} \cdot \mathrm{sqrt}\left(\frac{4\mathrm{e}0 \cdot \log(2\mathrm{e}0)}{\mathrm{Pi}}\right)\right);\ F := 0.5\mathrm{e}0 \cdot 1\mathrm{e}4$$

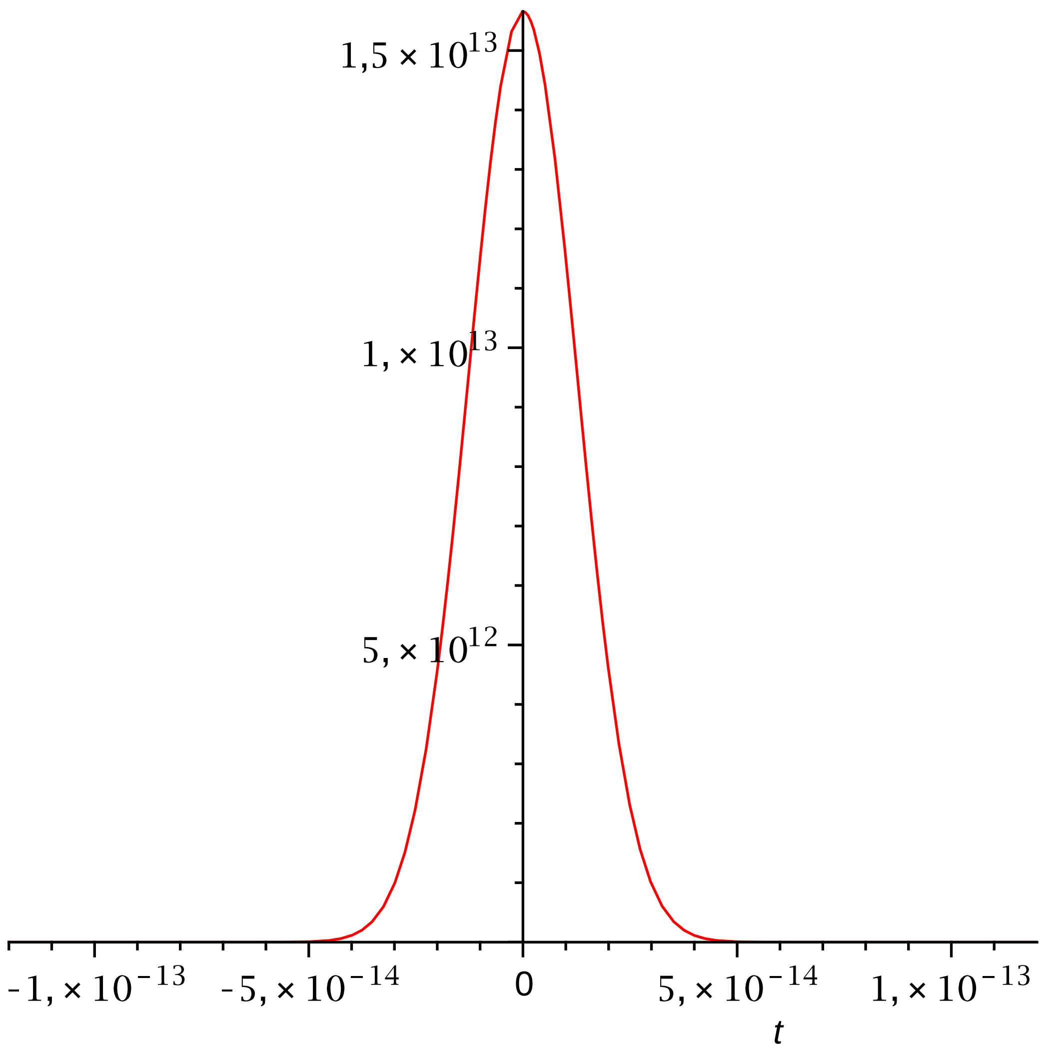
$$0.005039$$

$$1.329$$

$$3.1314575956655044458\ 10^{13}\ F$$

$$5000. \quad \textbf{(7)}$$

$$plot\left(Iabs(t) \cdot 1\mathrm{e}{-4},\ t = -4 \cdot \tau_{laser} .. 4 \cdot \tau_{laser}\right)$$



$evalf(MPI(t))$

$$8.3801175220676562503\ 10^{38}\left(\mathrm{e}^{-3.0806541358219791530\ 10^{27}\, t^{2.}}\right)^{5} \tag{8}$$

$eq1 := diff(ne(t),\, t) = C0 \cdot \exp(-C1 \cdot t^2)^k + \eta_{casc} \cdot ne(t)\# - g \cdot ne(t)\# - \eta_{rec} \cdot ne(t)^2$

$$\frac{\mathrm{d}}{\mathrm{d}t}\, ne(t) = C0\left(\mathrm{e}^{-C1\, t^2}\right)^5 + \eta_{casc}\, ne(t) \tag{9}$$

$neSol := dsolve\left(\left\{eq1,\ ne(-4 \cdot \tau_{laser}) = 0e0\right\}\right)$

$$ne(t) = \left(\frac{1}{10}\, \frac{C0\sqrt{\pi}\, \mathrm{e}^{\frac{1}{20}\frac{\eta_{casc}^2}{C1}}\sqrt{5}\, \mathrm{erf}\left(\sqrt{5}\sqrt{C1}\, t + \frac{1}{10}\, \frac{\eta_{casc}\sqrt{5}}{\sqrt{C1}}\right)}{\sqrt{C1}}\right. \tag{10}$$

$$- \frac{1}{10}\, \frac{1}{\sqrt{C1}}\left(C0\sqrt{\pi}\, \mathrm{e}^{\frac{1}{20}\frac{\eta_{casc}^2}{C1}}\sqrt{5}\, \mathrm{erf}\left(-\frac{3}{25000000000000}\sqrt{5}\sqrt{C1}\right.\right.$$

$$+\frac{1}{10}\,\frac{\eta_{casc}\sqrt{5}}{\sqrt{C1}}\Bigg)\Bigg)\Bigg)\,e^{\eta_{casc}\,t}$$

$$ne := t \to \left(\frac{1}{10}\,\frac{C0\sqrt{\pi}\,e^{\frac{1}{20}\,\frac{\eta_{casc}^2}{C1}}\sqrt{5}\,\mathrm{erf}\left(\sqrt{5}\,\sqrt{C1}\,t+\frac{1}{10}\,\frac{\eta_{casc}\sqrt{5}}{\sqrt{C1}}\right)}{\sqrt{C1}}\right.$$

$$\left.-\frac{1}{10}\,\frac{C0\sqrt{\pi}\,e^{\frac{1}{20}\,\frac{\eta_{casc}^2}{C1}}\sqrt{5}\,\mathrm{erf}\left(-\frac{3}{250000000000000}\sqrt{5}\,\sqrt{C1}+\frac{1}{10}\,\frac{\eta_{casc}\sqrt{5}}{\sqrt{C1}}\right)}{\sqrt{C1}}\right)$$

$$e^{\eta_{casc}\,t}$$

$$t \to \left(\frac{1}{10}\,\frac{C0\sqrt{\pi}\,e^{\frac{1}{20}\,\frac{\eta_{casc}^2}{C1}}\sqrt{5}\,\mathrm{erf}\left(\sqrt{5}\,\sqrt{C1}\,t+\frac{1}{10}\,\frac{\eta_{casc}\sqrt{5}}{\sqrt{C1}}\right)}{\sqrt{C1}}\right. \tag{11}$$

$$-\frac{1}{10}\,\frac{1}{\sqrt{C1}}\left(C0\sqrt{\pi}\,e^{\frac{1}{20}\,\frac{\eta_{casc}^2}{C1}}\sqrt{5}\,\mathrm{erf}\left(-\frac{3}{250000000000000}\sqrt{5}\,\sqrt{C1}\right.\right.$$

$$\left.\left.\left.+\frac{1}{10}\,\frac{\eta_{casc}\sqrt{5}}{\sqrt{C1}}\right)\right)\right)\,e^{\eta_{casc}\,t}$$

$\#C1 := 1.640585043\ 10^{26}$; $C0 := 1.711275329\ 10^{32}$;
$C0 := 8.3801175220676562503\ 10^{38}$; $C1 := 3.0806541358219791530\ 10^{27}$

$$8.3801175220676562503\ 10^{38}$$

$$3.0806541358219791530\ 10^{27} \tag{12}$$

τ_{laser}

$$3.0\ 10^{-14} \tag{13}$$

$ne(t)$

$$\Big(1.5098320416029157180\ 10^{24}\sqrt{\pi}\,e^{1.6230319210000838332\ 10^{-29}\,\eta_{casc}^2}\sqrt{5} \tag{14}$$

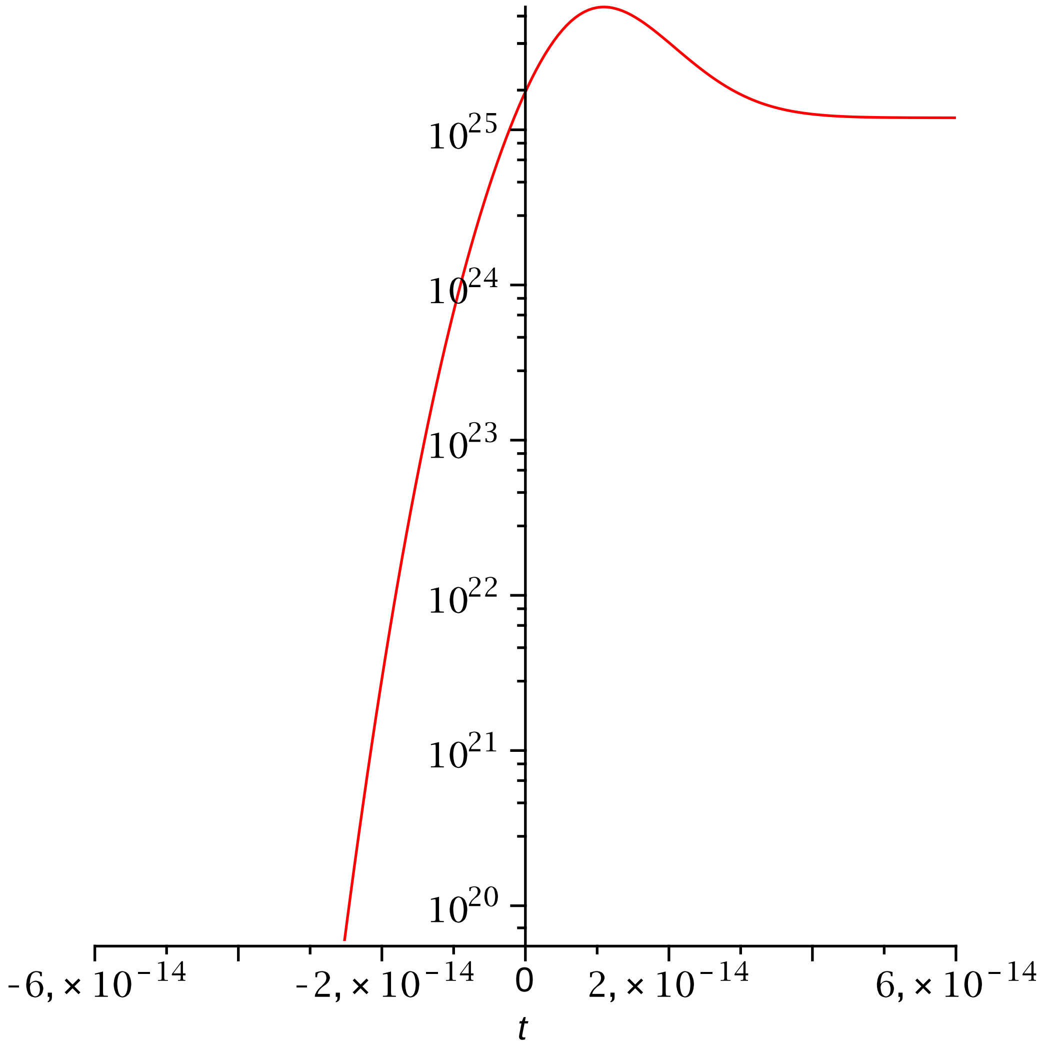
$$\mathrm{erf}\Big(5.5503640743846517091\ 10^{13}\sqrt{5}\,t$$

$$+1.8016836131796746923\ 10^{-15}\,\eta_{casc}\sqrt{5}\Big)$$

$$-1.5098320416029157180\ 10^{24}\sqrt{\pi}\,e^{1.6230319210000838332\ 10^{-29}\,\eta_{casc}^2}\sqrt{5}\,\mathrm{erf}\Big($$

$$-6.6604368892615820509\sqrt{5}+1.8016836131796746923\;10^{-15}\,\eta_{casc}\sqrt{5}\Big)\Big)$$
$$e^{\eta_{casc}\,t}$$

$with(plots) : logplot\left(ne(t),\ t=-2\cdot\tau_{laser}..2\cdot\tau_{laser},\ numpoints=200\right)$



Contribution of the MPI is low.

$evalf\,(ne(\mathrm{t}))$

$$\Big(-1.9148667344912161102\;10^{26}\,e^{3.7811626203541397287\;10^{-31}\left(2.4006136017076905415\;10^{15}\,e^{-3.0806541358219791530\;10^{27}\,t^{2.}}-1.6840012509358718144\;10^{11}\right)^{2}}\,\mathrm{erf}\Big(-1.2410991370196780443\;10^{14}\,t-1.4761651239109598271\,e^{-3.0806541358219791530\;10^{27}\,t^{2.}}+0.00010355118847471449463\Big) \qquad \textbf{(15)}$$

$+ 1.914866734491216110210^{26}$

$e^{3.781162620354139728710^{-31} (2.400613601707690541510^{15}}$

$e^{-3.080654135821979153010^{27} t^{2.}} - 1.684001250935871814410^{11})$

^{2} $\mathrm{erf}\big(14.893293195424611245$

$- 1.476165123910959827 1 e^{-3.080654135821979153010^{27} t^{2.}}\big)\big)$

$e^{0.15263324843958116171 (2.400613601707690541510^{15} e^{-3.080654135821979153010^{27} t^{2.}}}$

$- 1.684001250935871814410^{11}) t$

Considering cascade ionization.

$$\eta_{casc} := \frac{1e0}{\omega^2 \cdot \tau_{coll}^2 + 1e0} \cdot \left(\frac{ec^2 \cdot \tau_{coll}}{c \cdot n \cdot epsilon0 \cdot m \cdot Egap} \cdot Iabs(t) - \frac{m \cdot \omega^2 \cdot \tau_{coll}}{M} \right)$$

$$\frac{1}{0.56250000000000000000 \pi^2 + 1.} \big(1. \,(1.2003068008538452707\ 10^{15}$$
$$e^{-3.080654135821979153010^{27} t^{2.}} - 1.7062500000000000000\ 10^{10} \pi^2)\big) \quad \textbf{(16)}$$

$$g := 0e0\# \frac{\tau_{coll} \cdot Egap}{3e0 \cdot m} \cdot \left(\left(\frac{2.4e0}{omega0} \right)^2 + \left(\frac{1e0}{zr} \right)^2 \right); omega0 := 5e-6; zr := \frac{\mathrm{Pi} \cdot \omega 0^2}{\left(\frac{\mathrm{lambda}}{n} \right)}$$

$$0. \quad \textbf{(17)}$$

$restart:$

$$r := \frac{(\mathrm{sqrt}(\varepsilon_1) - \mathrm{sqrt}(\varepsilon_2))}{\mathrm{sqrt}(\varepsilon_1) + \mathrm{sqrt}(\varepsilon_2)}; R := (\mathrm{abs}(r))^2; \varepsilon_1 := 1e0; \varepsilon_2 := 1.329^2$$

$$-0.1412623444$$

$$0.01995504995$$

$$1.$$

$$1.766241 \quad \textbf{(18)}$$